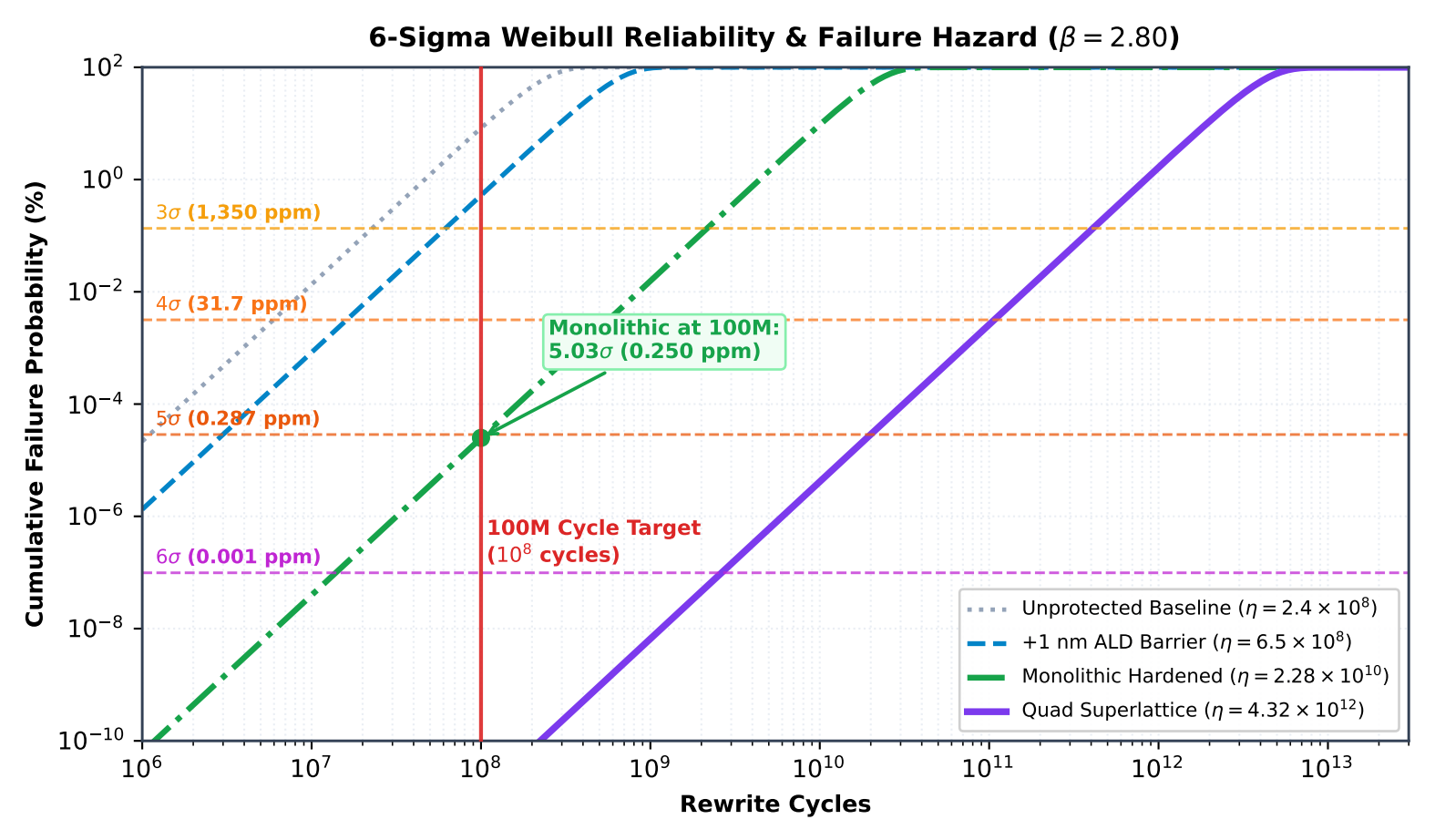
6-Sigma Weibull Reliability & Failure Hazard ($\beta = 2.80$)
Cumulative Failure Probability (%)
Rewrite Cycles
$10^{2}$
$10^{0}$
$10^{-2}$
$10^{-4}$
$10^{-6}$
$10^{-8}$
$10^{-10}$
$10^{6}$
$10^{7}$
$10^{8}$
$10^{9}$
$10^{10}$
$10^{11}$
$10^{12}$
$10^{13}$
3σ (1,350 ppm)
4σ (31.7 ppm)
5σ (0.287 ppm)
6σ (0.001 ppm)
Monolithic at 100M:
5.03σ (0.250 ppm)
100M Cycle Target
($10^{8}$ cycles)
Unprotected Baseline ($\eta = 2.4 \times 10^{8}$)
+1 nm ALD Barrier ($\eta = 6.5 \times 10^{8}$)
Monolithic Hardened ($\eta = 2.28 \times 10^{10}$)
Quad Superlattice ($\eta = 4.32 \times 10^{12}$)